

Packet Switching Delay – Student Worksheet

Course: Data Communications & Computer Networks

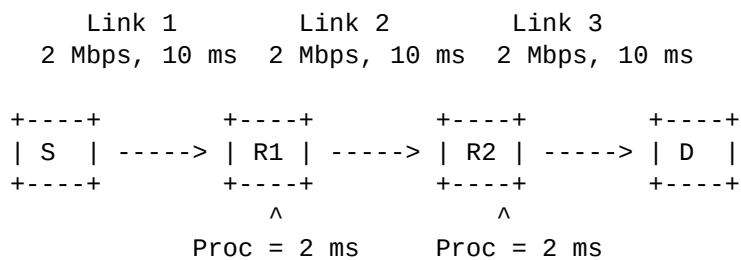
Problem Statement

A source node **S** wants to send a message of size **6000 bytes** to a destination node **D** using **packet switching**. The message is divided into **3 equal-sized packets**. The packets travel through two intermediate routers (**R1** and **R2**) before reaching the destination.

Network Parameters

- Link data rate: 2 Mbps (all links)
- Propagation delay per link: 10 ms
- Processing delay at each router: 2 ms
- Queueing delay: Negligible
- Packet headers: Ignored
- Switching technique: Store-and-forward packet switching

Network Diagram



Student Tasks

1. Determine the size of each packet in bytes and bits.
2. Calculate the transmission delay for one packet on one link.
3. Calculate the end-to-end delay for the first packet.
4. Calculate the total time required for all packets.
5. Briefly explain why packet switching is efficient in this scenario.

Instructor Solution

Step 1: Packet Size

Total message size = 6000 bytes, Number of packets = 3. Each packet size = 2000 bytes = 16,000 bits.

Step 2: Transmission Delay

Transmission Delay = Packet size / Link rate = $16,000 / (2 \times 10^6) = 0.008 \text{ s} = 8 \text{ ms}$.

Step 3: End-to-End Delay (First Packet)

Each link contributes 8 ms transmission + 10 ms propagation. There are 3 links and 2 routers with 2 ms processing delay each. Total delay = $3 \times (8 + 10) + 2 \times 2 = 58 \text{ ms}$.

Step 4: Total Time for All Packets

Due to pipelining, additional packets add only one transmission delay each. Total time = $58 + (3 - 1) \times 8 = 74 \text{ ms}$.

Conceptual Explanation

Packet switching allows multiple packets to be in transit simultaneously on different links. This pipelining improves link utilization and reduces overall transmission time compared to circuit switching.